

sovereign-validation-loop (v6.0) 🦅

An enterprise-grade, deterministic CLI test harness engineered for local LLM dual-model orchestration natively inside your RAM. This suite runs an adversarial **Generator-Validator pipeline**, parses feedback with a tolerant scoring engine, and forces surgical VRAM purging via ``keep_alive => 0`` to prevent system-throttling page swaps and port collisions.

🛠️ Advanced Engineering Features

Version 6.0 completely rewrites the pipeline architecture to ensure absolute stability over heavy testing runs:

- Surgical VRAM Isolation:** Queries the local ``/api/ps`` registry using active hooks after every complete test case, forcibly unloading residual streams to prevent memory-stack leaks.

- Resumable Execution Matrix:** Automatically scans your existing CSV logging logs on boot. If an overnight operation is interrupted, it skips completed lines and appends new records seamlessly without losing data history.

- Context Window Protection:** Hardcodes an explicit ``num_ctx => 8192`` boundary to prevent local API runtimes from silently truncating long text strings or complex papers.

- Tolerant Regex Score Parsing:** Utilizes a fallback matrix of four distinct regular expression patterns to accurately extract scores from alternative layout styles (e.g., markdown bolding or fractions).

- Automated Key Denylisting:** Automatically filters out non-chat configurations (like embedding or reranker models) to preserve execution flow across your directory.

📊 Telemetry Log Output Matrix

The execution pipeline generates a highly detailed, real-time data ledger (``benchmark_results.csv``) mapping twelve distinct tracking metrics:

Timestamp	Generator	Validator	Lens	Status	Final_Score	Score_Parsed	Loops	Duration
2026-09-16 12:00:30	deepseek-r1:14b	dolphin3:latest	objectivity	PASS	100	YES	1	317.25s
2026-09-16 16:09:05	qwen3-abliterated:4b	dolphin3:latest	objectivity	PASS	100	YES	1	43.99s

🚀 Quick Start Deployment

1. System Requirements

- PHP 8.2+ CLI** environment enabled locally (e.g., via XAMPP).

- Ollama** running natively on default host port ``11434``.

- For full asynchronous interrupt safety, the ``pcntl`` extension is highly recommended.

2. Prepare the Environment

Ensure your local VRAM boundaries are protected by removing un-optimized blocks and fetching high-speed, edge-ready weights:

```
```bash
Clear out un-optimized, resource-heavy experimental variants
ollama rm schroneko/mistral-nemo-minitron-8b-instruct:latest

Fetch the high-speed, edge-optimized 3B Western logic layer
ollama run llama3.2:3b
```
```

3. Execution Commands

Navigate to your local working directory inside your terminal console and run your targeted processing routines:

```
```bash
Run the complete diagnostic pairing matrix from scratch
php bench_mark.php

Resume an interrupted overnight session from the exact line it stopped
php bench_mark.php --resume

Isolate testing routines to specific models within your active directory
php bench_mark.php --models=deepseek-r1:14b,llama3.2:3b

Run a safe configuration check without initializing the model weights
php bench_mark.php --dry-run
```
```

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